

Exploring Aligned and Disaligned Multimodal Features in Recommender Systems

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Outline

- 1 Introduction
- 2 Aligned Models
- 3 Experimental Settings
- 4 Experiments
- 5 Discussion & Conclusion

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Recommender Systems

- Guide users to relevant items in a large space [5]
- Approaches: Content-Based, Collaborative, Hybrid, Knowledge-Based, Context-Aware

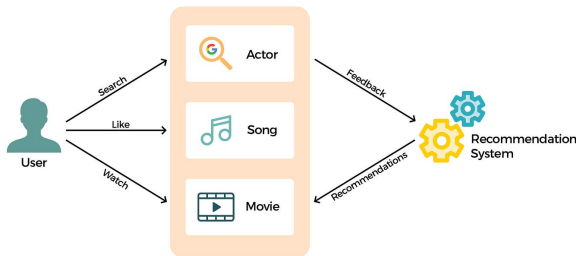


Figure: Example of a Recommender System

Multimodal Recommender Systems

- Use multiple modalities (text, image, audio) to enrich item representations [7]
- Encoding options: concat, fusion/attention, GNNs [1, 4]

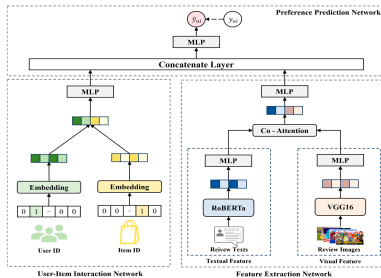


Figure: Example of a Multimodal Recommender System

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Contrastive Learning

- Self-supervised method learning representations that bring similar samples closer [3]
- Contrastive loss: $\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} [k \neq i] \exp(\text{sim}(z_i, z_k)/\tau)}$

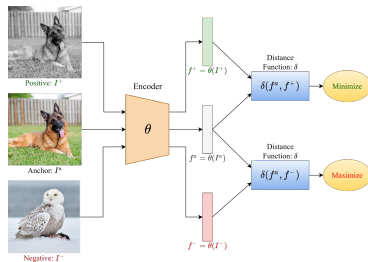


Figure: Conceptual representation of Contrastive Learning

- CLIP aligns image and text embeddings in a shared space (OpenAI) [6]
- Recent work shows that large multimodal encoders like CLIP can effectively improve recommender system performance when fine-tuned appropriately [8]

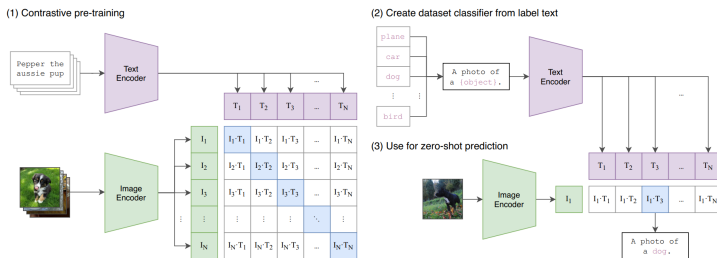


Figure: Conceptual representation of CLIP architecture

AudioCLIP

- Extends CLIP by adding an audio encoder (ESResNeXt) to the tri-modal alignment [2]
- Training: audio-head pretrain, cooperative distillation, full tri-modal fine-tuning on AudioSet

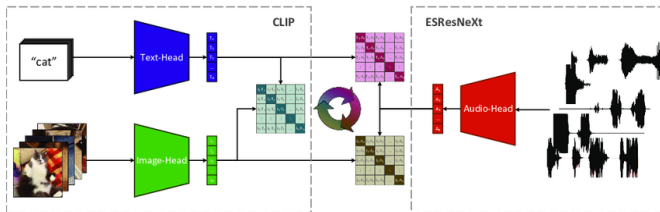


Figure: Conceptual representation of AudioCLIP architecture

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RQ1: Can aligned multimodal embeddings improve recommender performance?

- MovieLens 1M: movies with posters, trailers (audio), and text descriptions
- LastFM: artist-level interactions, track-level multimodal features (images, audio, text)

Outline

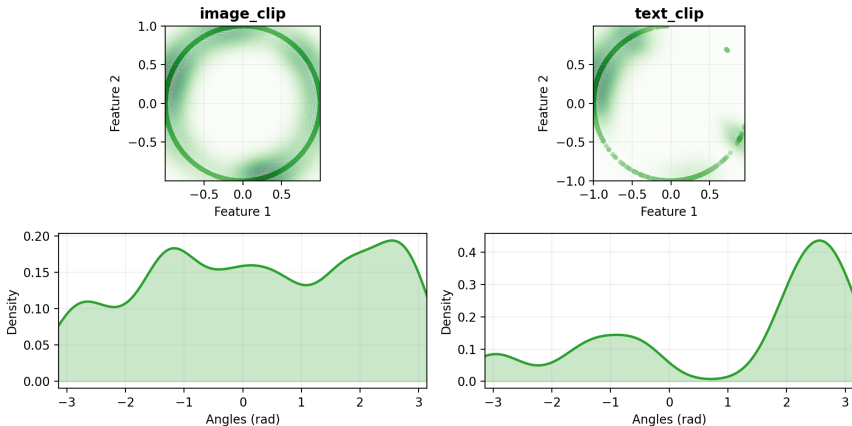
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Embedding distribution technique

- UMAP (Uniform Manifold Approximation and Projection) is used to reduce high-dimensional embeddings to 2D preserving local/global structure.
- After UMAP, we center and S1-normalize points: $\mathbf{x}'_i = \mathbf{x}_i - \bar{\mathbf{x}}$,
 $\mathbf{x}''_i = \frac{\mathbf{x}'_i}{\|\mathbf{x}'_i\|}$.
- Spatial density: 2D Gaussian Kernel Density Estimation (KDE) on the projected points.

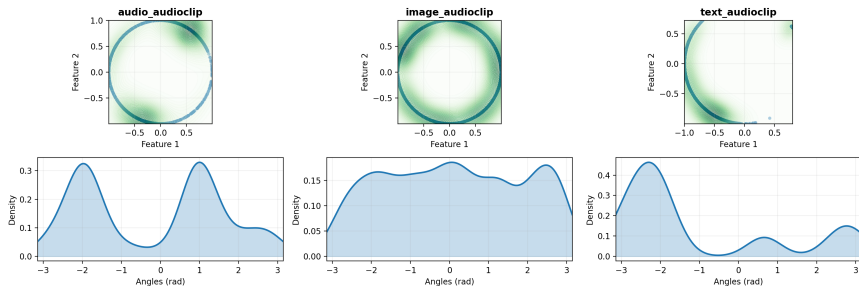
MovieLens 1M: CLIP-aligned Text+Image

MOVIELENS_1M - text_clip_image_clip



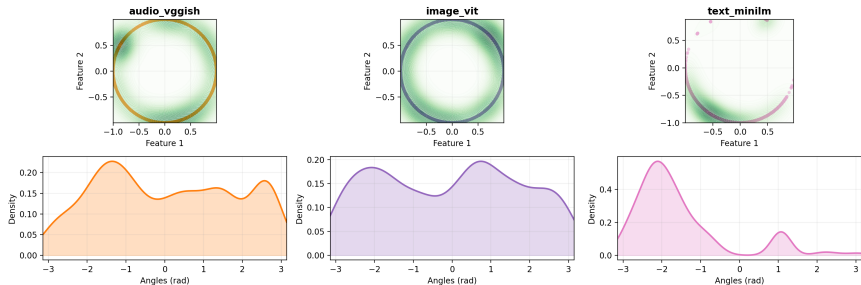
MovieLens 1M: AudioCLIP aligned (Text+Image+Audio)

MOVIELENS_1M - audioclip_full



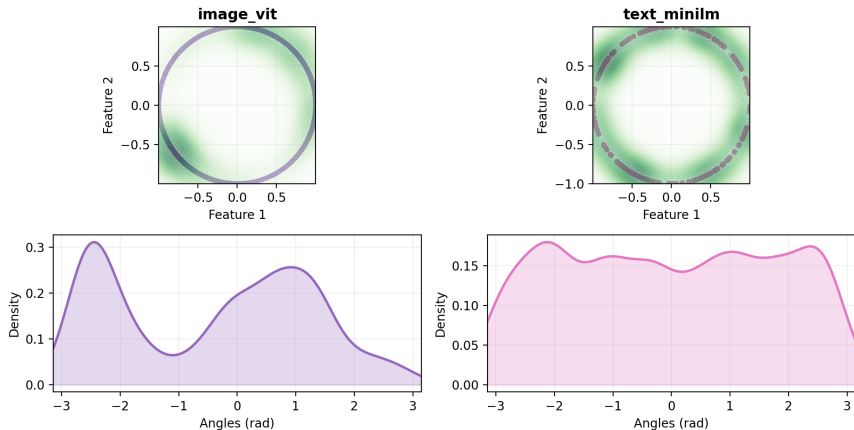
MovieLens 1M: Disaligned Text+Image+Audio (MiniLM+ViT+VGGish)

MOVIELENS_1M - text_minilm_image_vit_audio_vggish



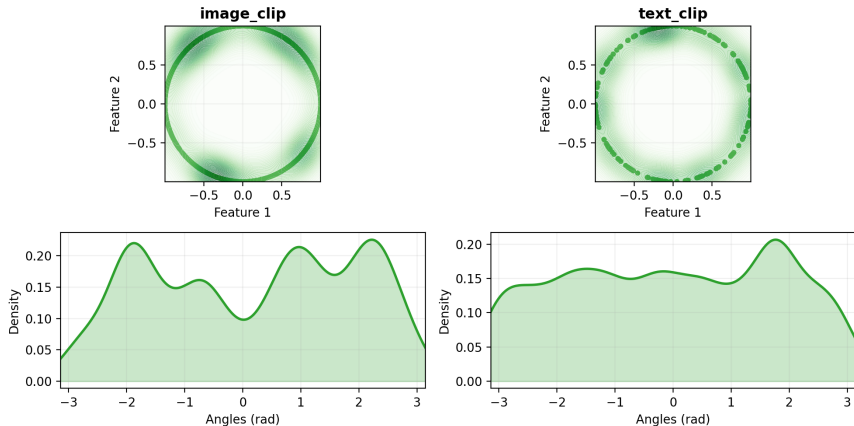
LastFM: Disaligned Text+Image (MiniLM + ViT)

LASTFM - text_minilm_image_vit



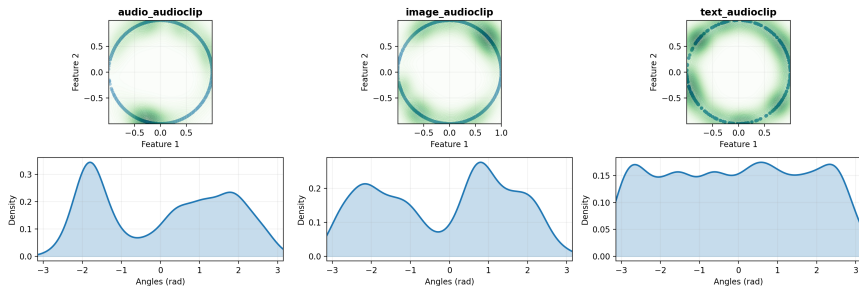
LastFM: CLIP Text+Image aligned

LASTFM - text_clip_image_clip



LastFM: AudioCLIP aligned (Text+Image+Audio)

LASTFM - audioclip_full



Quantitative comparison: MovieLens (k=5)

Model	rec@5	ndcg@5
VBPR (Text - MiniLM)	0.1144	0.2196
LATTICE_1 (Text - MiniLM)	0.1131	0.2169
FREEDOM (Text - MiniLM)	0.1133	0.2173

Table: MovieLens 1M - Comparison at cutoff k=5 (Text features)

Quantitative comparison: MovieLens (k=10)

Model	rec@10	ndcg@10
VBPR (Text - MiniLM)	0.1877	0.2269
LATTICE_1 (Text - MiniLM)	0.1848	0.2238
FREEDOM (Text - MiniLM)	0.1836	0.2238

Table: MovieLens 1M - Comparison at cutoff k=10 (Text features)

Quantitative comparison: LastFM (k=5)

Model	rec@5	ndcg@5
VBPR (Text - MiniLM)	0.0733	0.1763
LATTICE_1 (Text - MiniLM)	0.0679	0.1657
FREEDOM (Text - MiniLM)	0.0522	0.1358

Table: LastFM - Comparison at cutoff k=5 (Text features)

Quantitative comparison: LastFM (k=10)

Model	rec@10	ndcg@10
VBPR (Text - MiniLM)	0.1268	0.1695
LATTICE_1 (Text - MiniLM)	0.1222	0.1627
FREEDOM (Text - MiniLM)	0.0988	0.1321

Table: LastFM - Comparison at cutoff k=10 (Text features)

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- Alignment helps when modalities share strong semantic relationships (e.g., posters and descriptions)
- Aggressive alignment (AudioCLIP) can erase modality-specific signals leading to worse ranking
- Simpler models (VBPR) can outperform complex GNNs when features are well represented

Conclusions and Future Work

- Use domain-adaptive fusion strategies to select aligned/disaligned features
- Explore softer alignment / regularization strategies for audio
- Investigate model-feature co-design (when to use GNNs vs. projection models)

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